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(54) **ISOLATION TRANSFORMER PACKAGES WITH MAGNETOSTRICTION MANAGEMENT**

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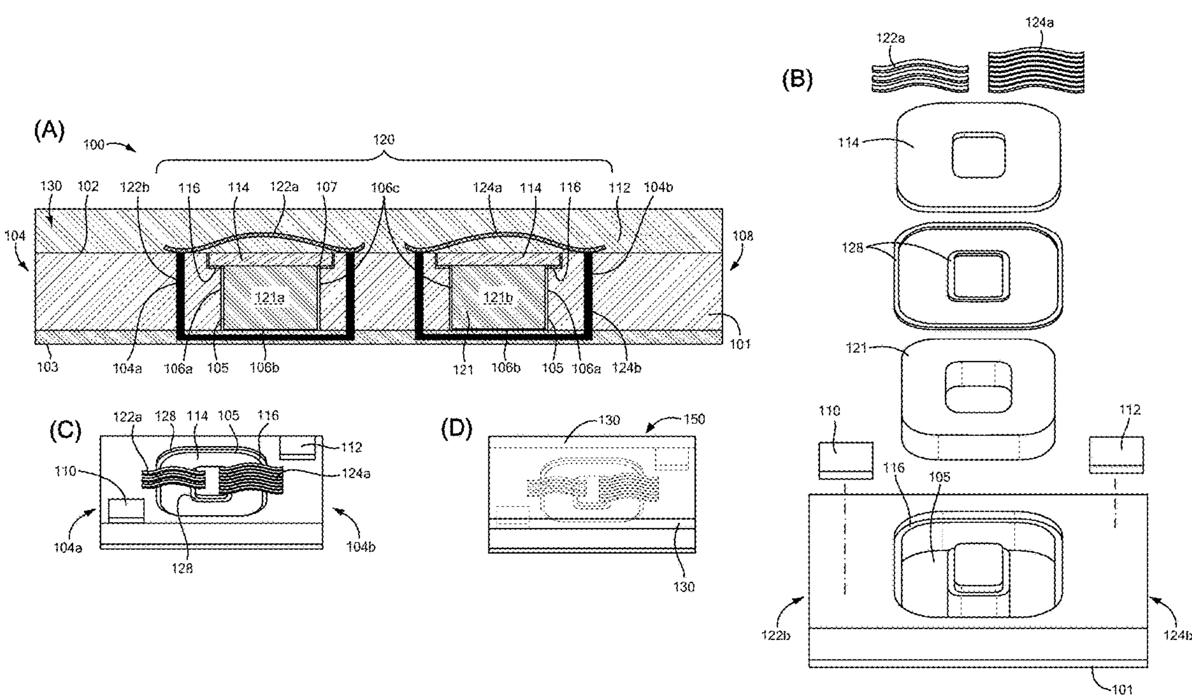
(57) **ABSTRACT**

Isolation transformer packages and structures and related methods reduce or minimize deleterious effects arising from magnetostriction during operation of the included transformer. An example transformer based integrated circuit package includes a substrate including a cavity, with the cavity including an aperture. A magnetic core is disposed in the cavity, with the magnetic core includes a soft ferromagnetic material. The cavity is configured to provide a space between an interior surface of the cavity and an exterior surface of the magnetic core. A cap is disposed in the aperture and configured to seal the aperture. A plurality of conductive traces forming first and second coils is disposed about the magnetic core, with the first and second coils and magnetic core forming a transformer.



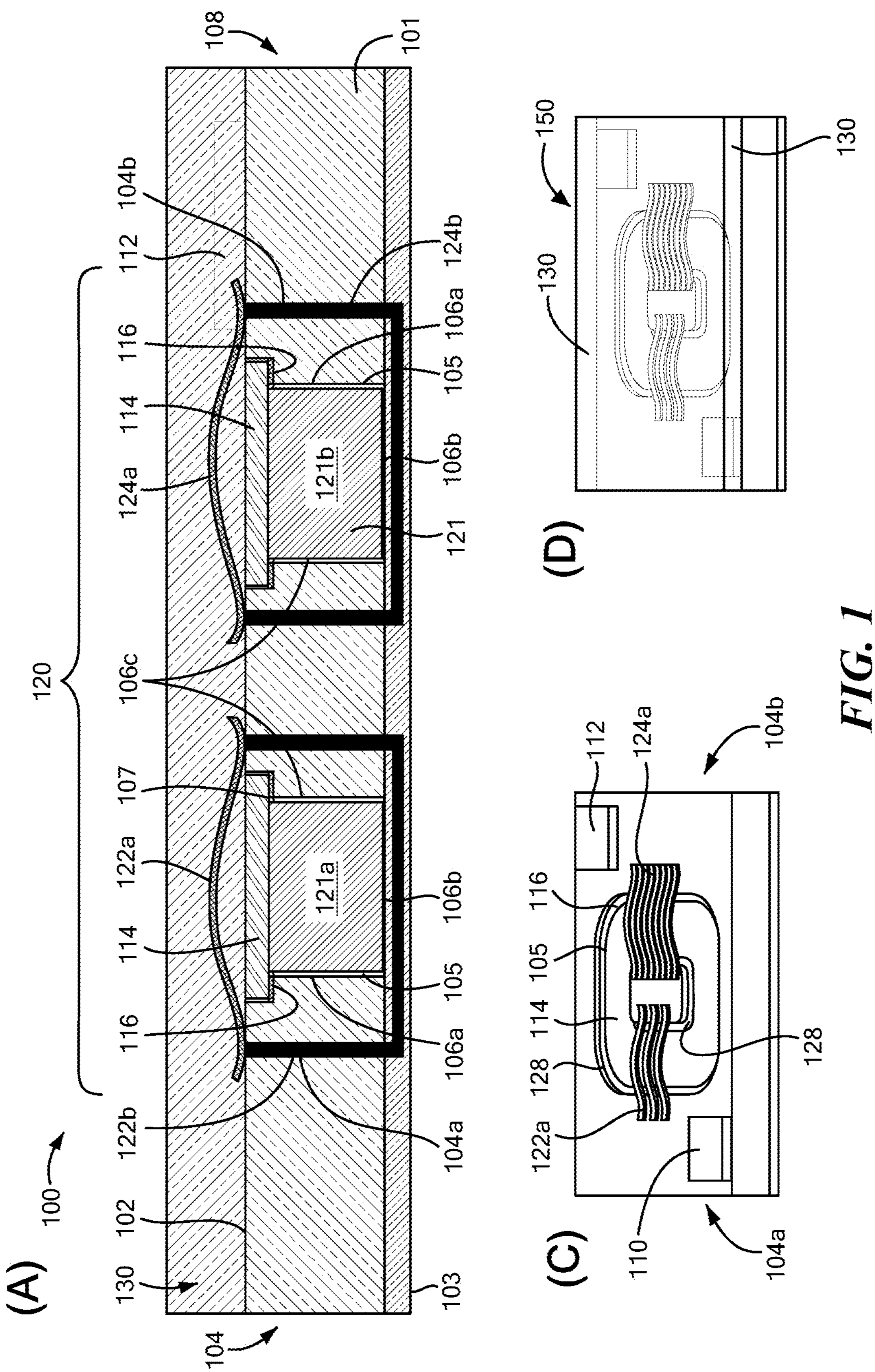

(A)
100
102
104
108
101
103
130
120
104a
104b
105
106a
106b
106c
107
112
114
116
121
121a
121b
122a
122b
124a
124b
(C)
104a
104b
110
112
114
116
105
128
122a
124a
(D)
150
130

*FIG. 1*

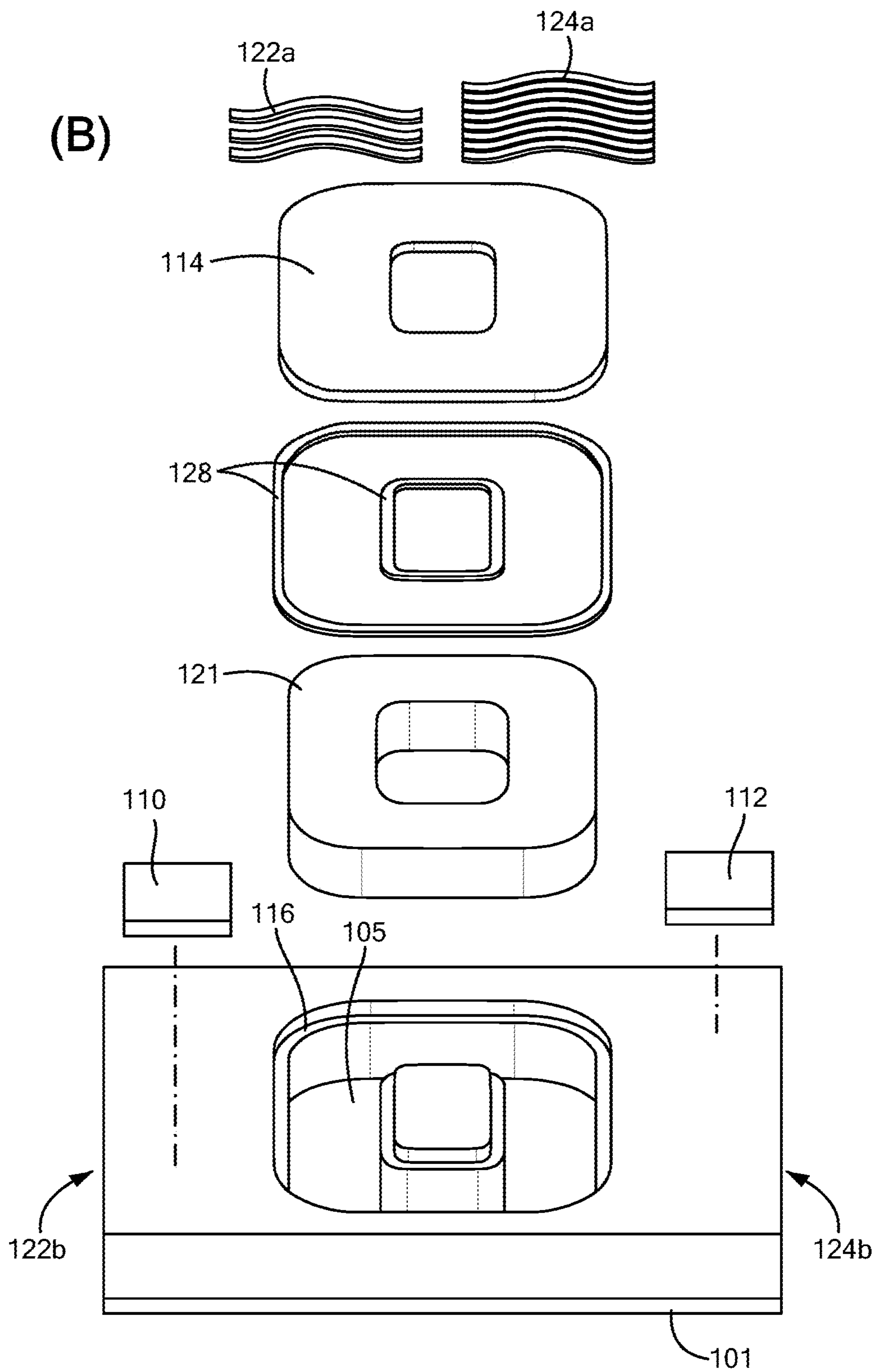



*FIG. 1*

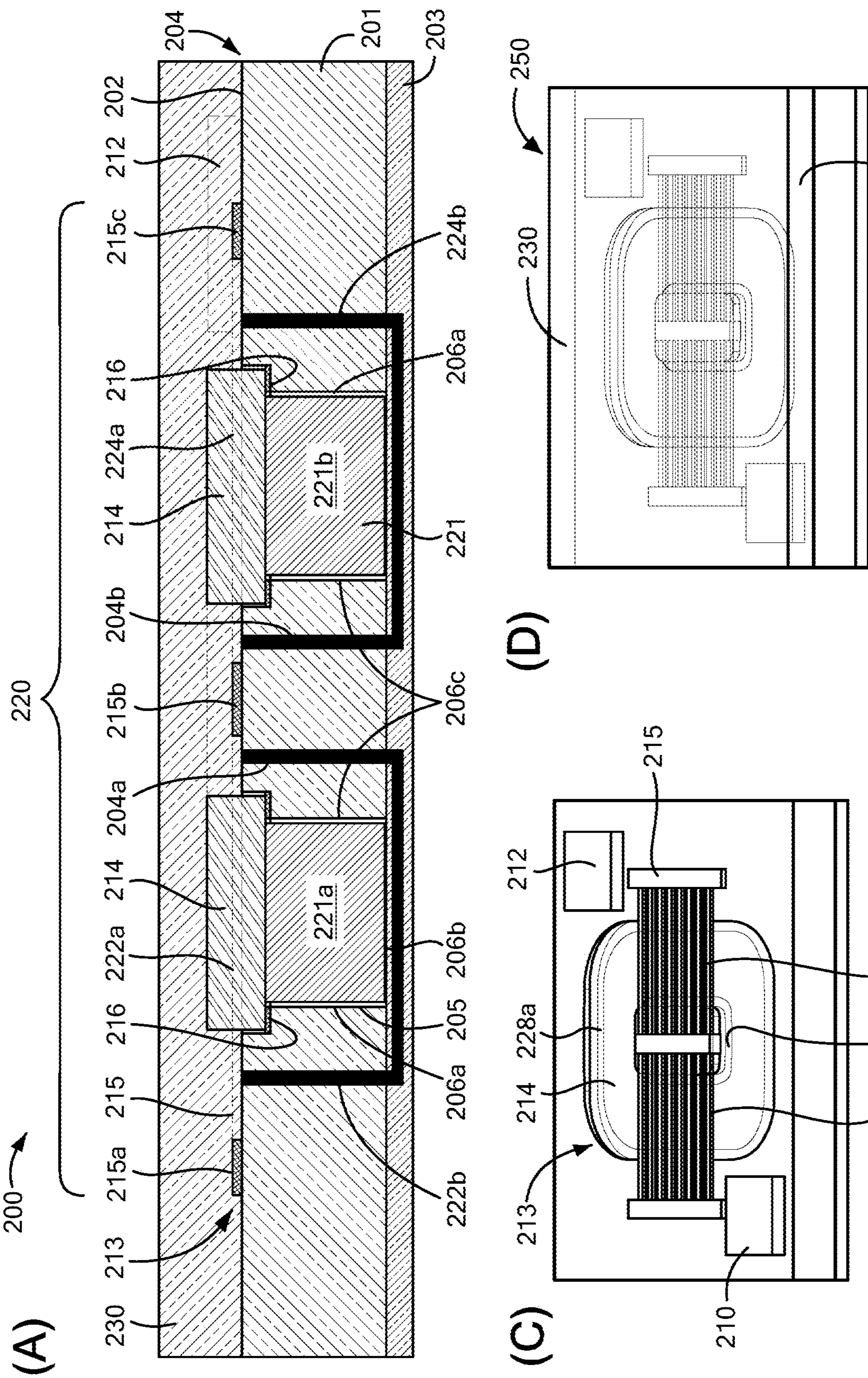

(A)
200
220
230
213
215a
215
216
222a
214
204a
215b
204b
214
224a
216
215c
212
202
204
201
203
224b
206a
221b
221
206c
221a
206b
205
206a
222b
(C)
210
213
214
228a
212
215
224a
228b
222a
(D)
250
230
230

*FIG. 2*

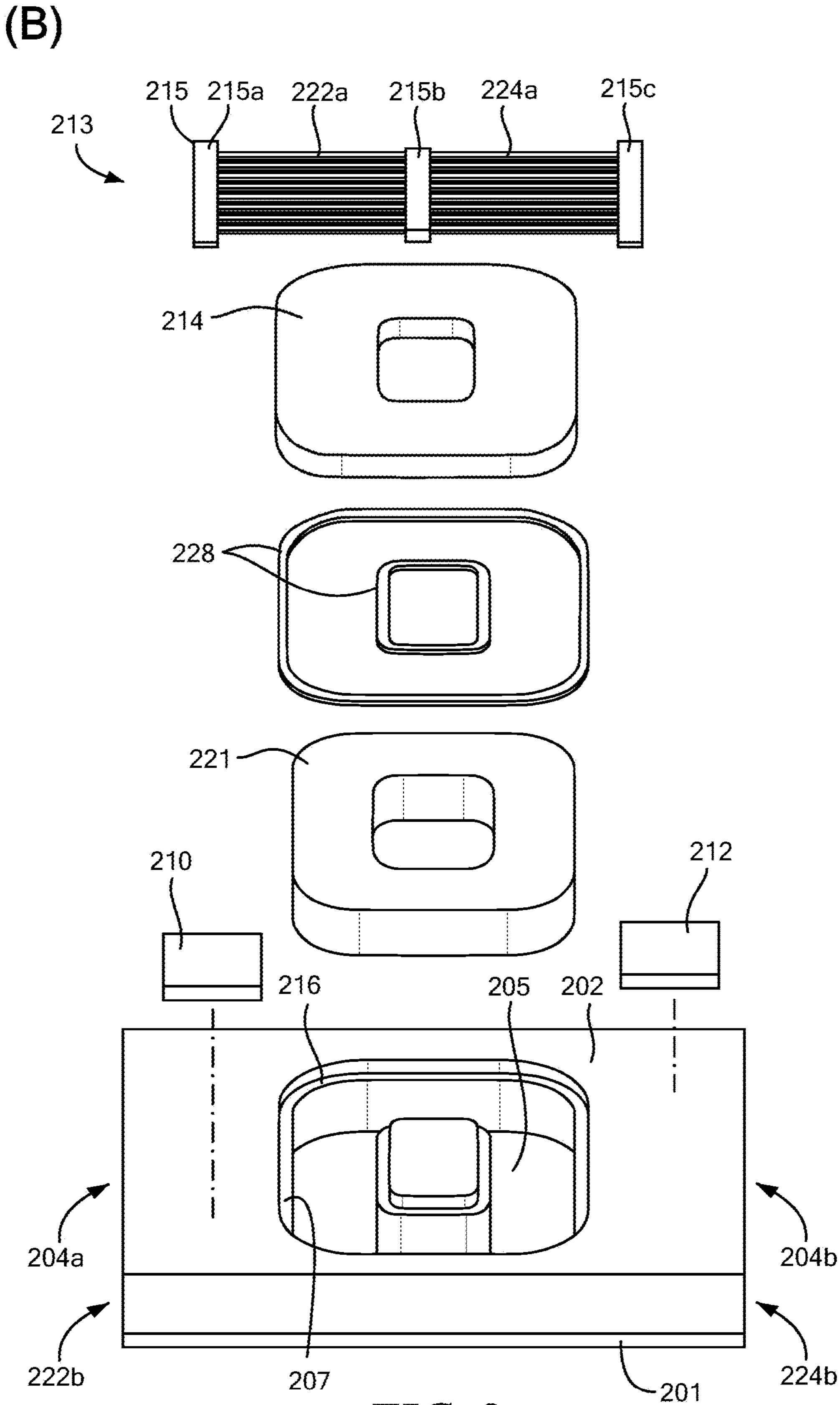

(B)
213
215
215a
222a
215b
224a
215c
214
228
221
210
212
216
205
202
204a
204b
222b
224b
207
201

FIG. 2

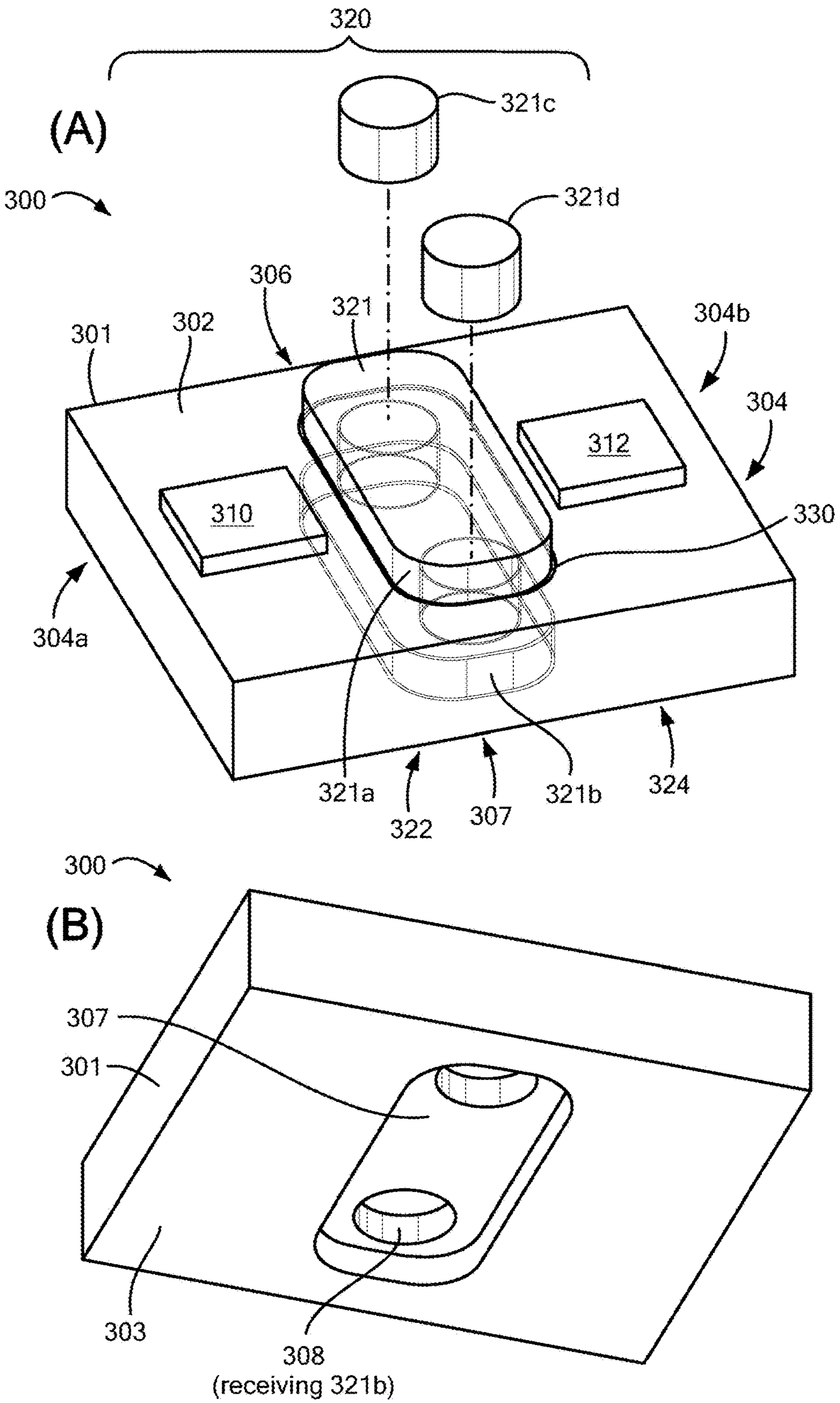

320
321c
(A)
300
321d
306
321
301
302
304b
304
312
310
330
304a
321a
322
307
321b
324
300
(B)
307
301
303
308
(receiving 321b)

*FIG. 3*

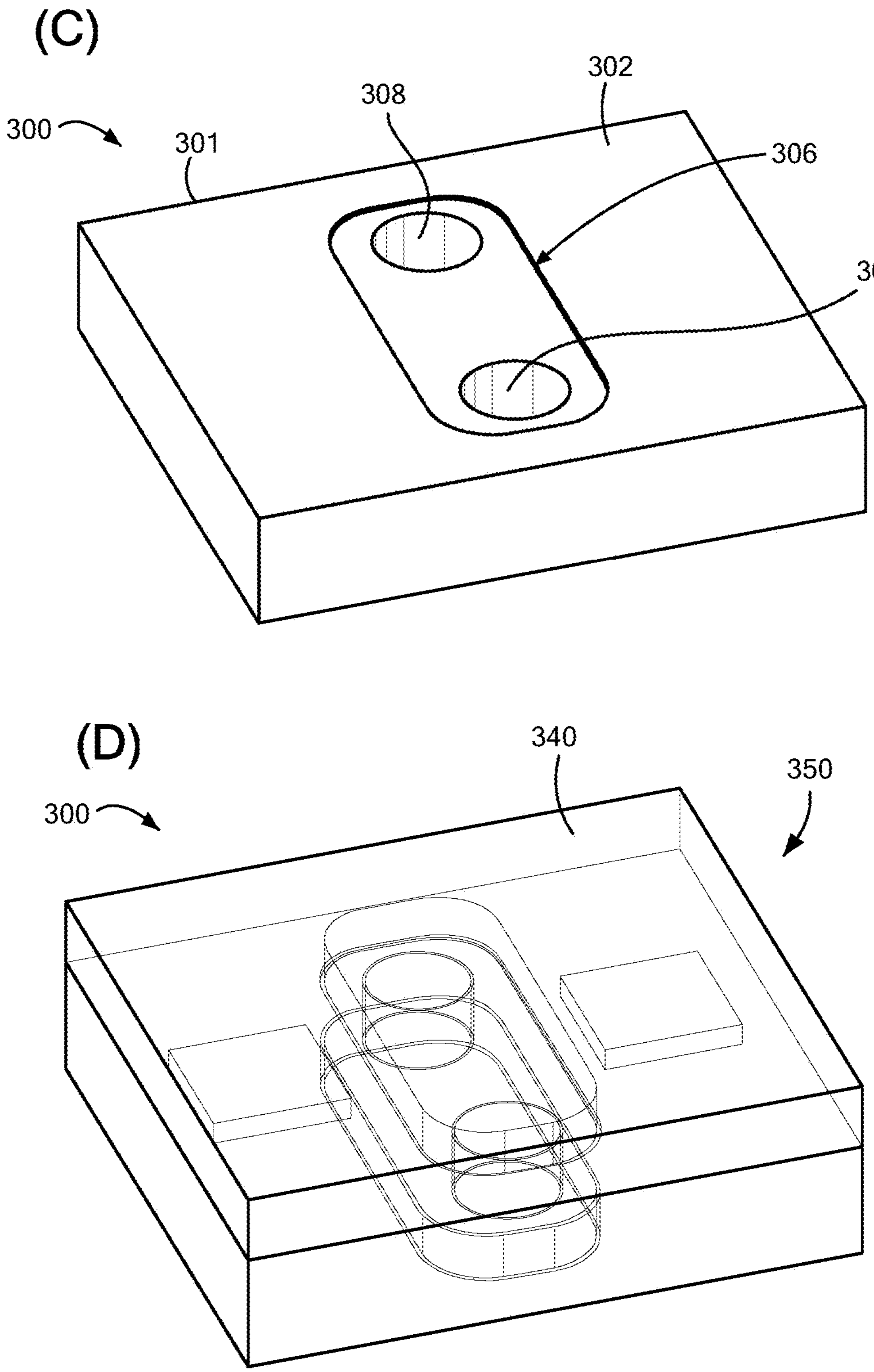



*FIG. 3*

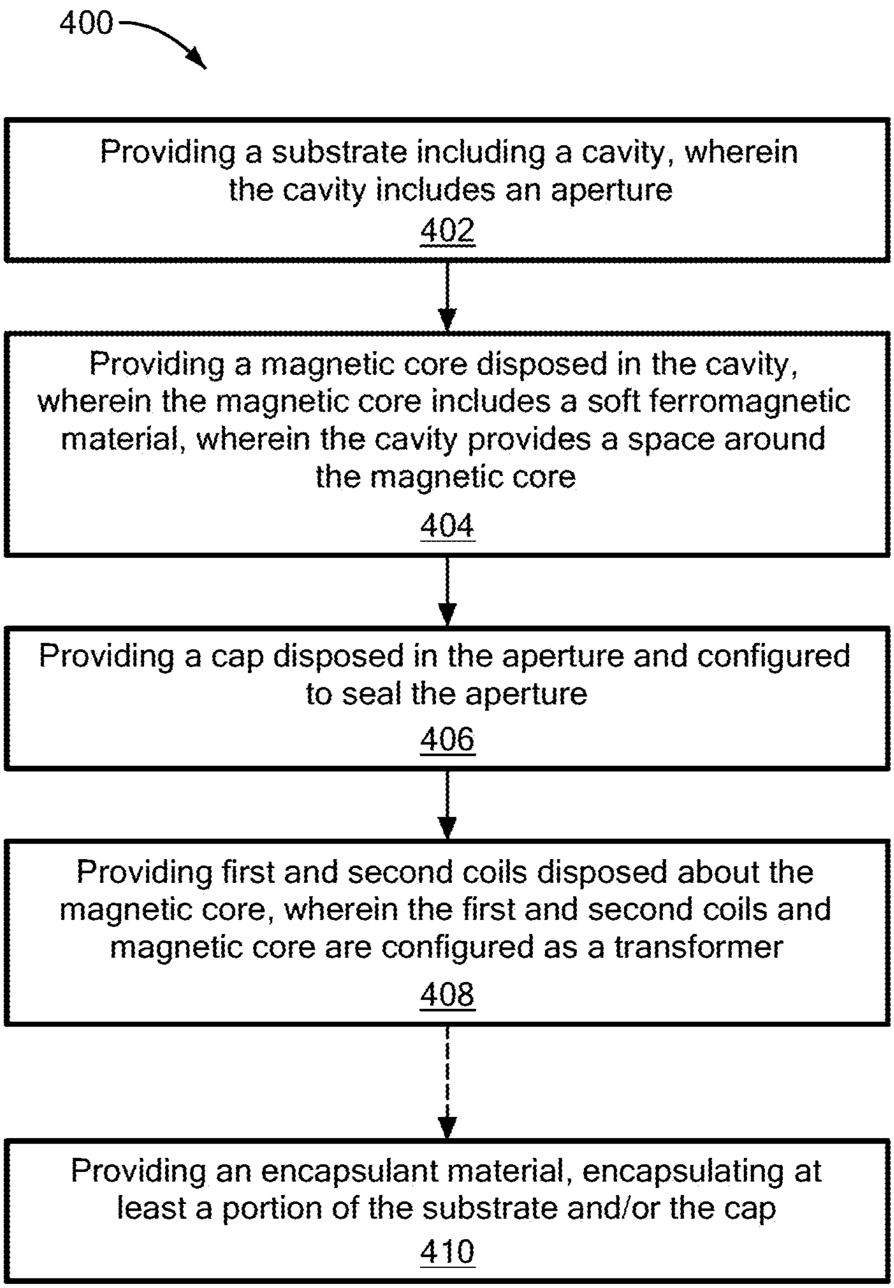

400
Providing a substrate including a cavity, wherein the cavity includes an aperture
402
Providing a magnetic core disposed in the cavity, wherein the magnetic core includes a soft ferromagnetic material, wherein the cavity provides a space around the magnetic core
404
Providing a cap disposed in the aperture and configured to seal the aperture
406
Providing first and second coils disposed about the magnetic core, wherein the first and second coils and magnetic core are configured as a transformer
408
Providing an encapsulant material, encapsulating at least a portion of the substrate and/or the cap
410

*FIG. 4*

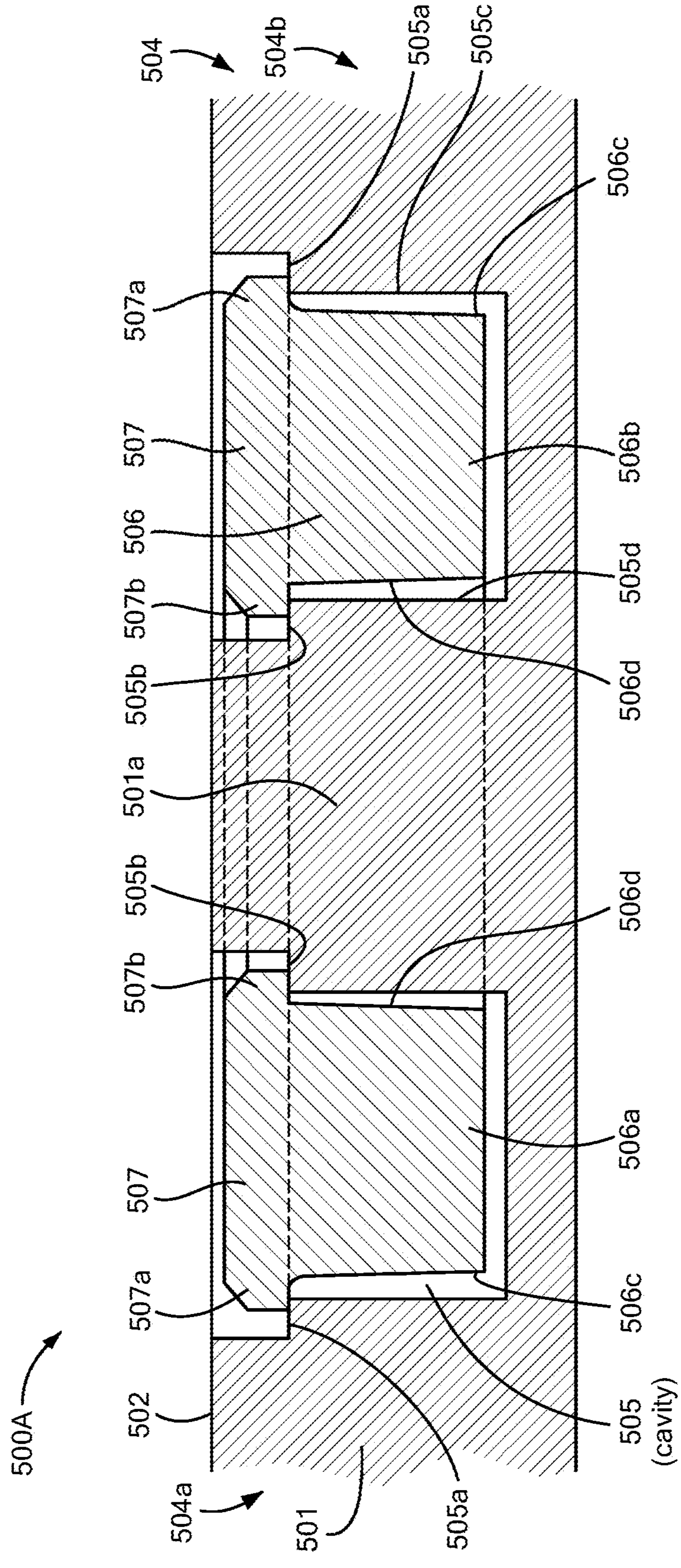

504
504b
505a
505c
506c
507a
507
506
506b
507b
505d
505b
506d
501a
505b
506d
507b
507
506a
507a
506c
500A
502
505
(cavity)
504a
501
505a

*FIG. 5A*

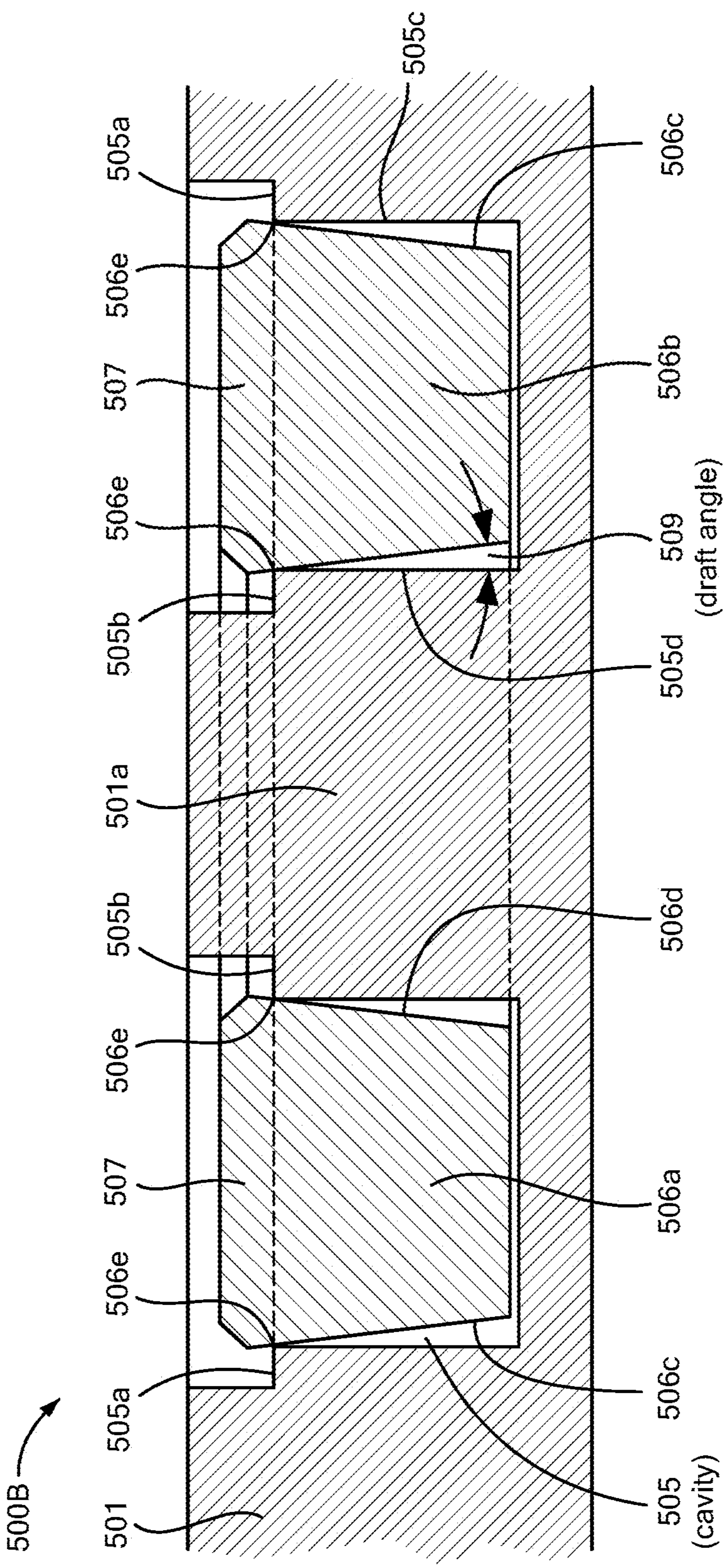

500B
501
505a
506e
507
506e
505b
501a
505b
506e
507
506e
505a
505c
506c
506b
509
(draft angle)
505d
506d
506a
506c
505
(cavity)

FIG. 5B

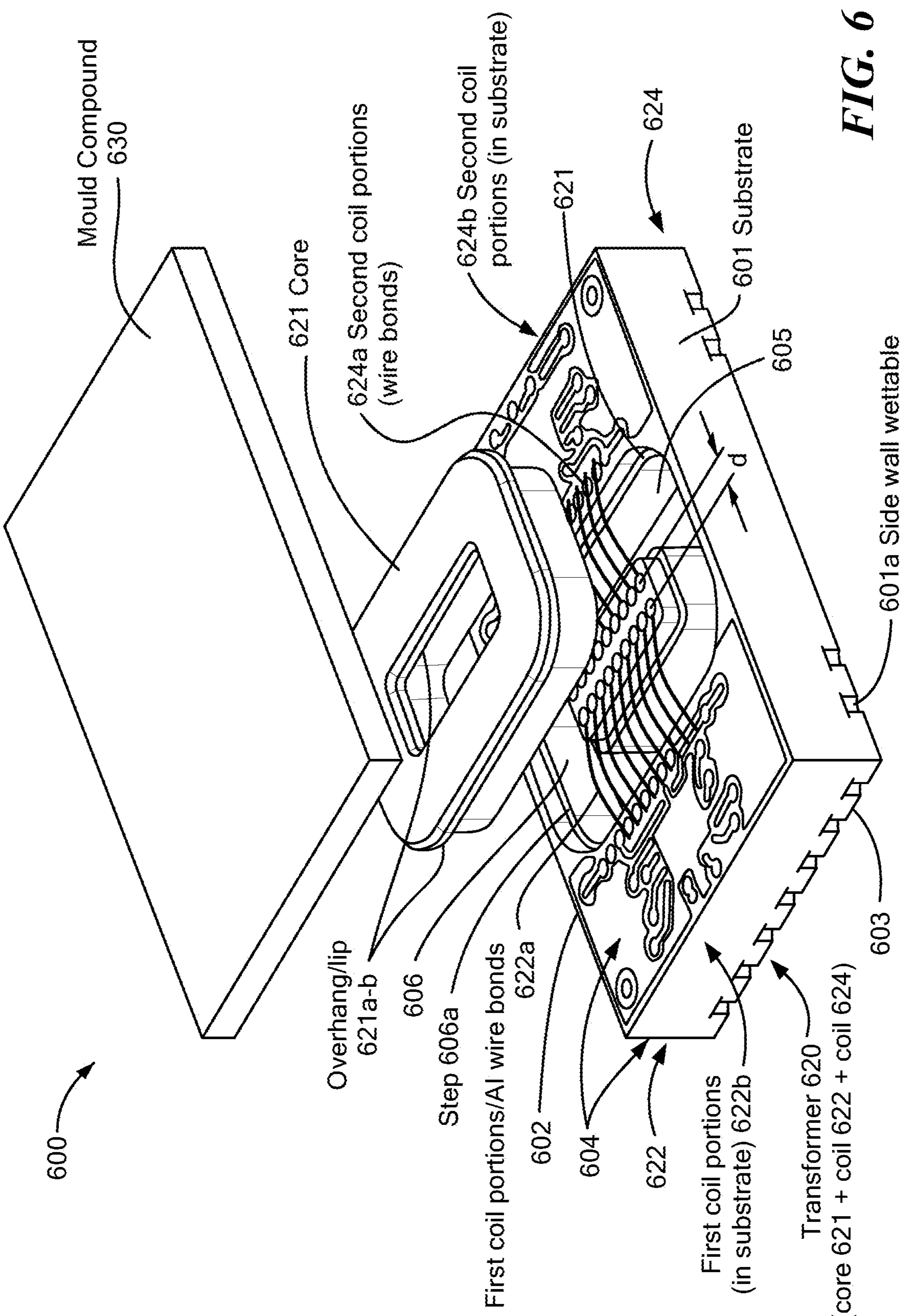

600
Mould Compound
630
621 Core
624a Second coil portions
(wire bonds)
624b Second coil
portions (in substrate)
621
624
601 Substrate
605
d
601a Side wall wettable
603
Overhang/lip
621a-b
606
Step 606a
First coil portions/Al wire bonds
622a
602
604
622
First coil portions
(in substrate) 622b
Transformer 620
(core 621 + coil 622 + coil 624)

FIG. 6

# ISOLATION TRANSFORMER PACKAGES WITH MAGNETOSTRICTION MANAGEMENT

## BACKGROUND

**[0001]** Solid state switches typically include a transistor structure. The controlling electrode of the switch, usually referred to as its gate (or base), is typically controlled (driven) by a switch drive circuit, sometimes also referred to as gate drive circuit. Such solid state switches are typically voltage-controlled, turning on when the gate voltage exceeds a manufacturer-specific threshold voltage by a margin, and turning off when the gate voltage remains below the threshold voltage by a margin.

**[0002]** Switch drive circuits typically receive their control instructions from a controller such as a pulse-width-modulated (PWM) controller via one or more switch driver inputs. Switch drive circuits deliver their drive signals directly (or indirectly via networks of active and passive components) to the respective terminals of the switch (gate and source).

**[0003]** Some electronic systems, including ones with solid state switches, have employed galvanic isolation to prevent undesirable DC currents flowing from one side of an isolation barrier to the other. Such galvanic isolation can be used to separate circuits in order to protect users from coming into direct contact with hazardous voltages.

**[0004]** Various transmission techniques are available for signals to be sent across galvanic isolation barriers including optical, capacitive, and magnetic coupling techniques. Magnetic coupling typically relies on use of a transformer to magnetically couple circuits on the different sides of the transformer, typically referred to as the primary and secondary sides, while also providing galvanic separation of the circuits,

**[0005]** Transformers used for magnetic-coupling isolation barriers typically utilize a magnetic core to provide a magnetic path to channel flux created by the currents flowing in the primary and secondary sides of the transformer. Magnetic-coupling isolation barriers have been shown to have various drawbacks, including manufacturing problems, for integrated circuit (IC) packages due to the included magnetic core.

## SUMMARY

**[0006]** Aspects of the present disclosure are directed to isolation transformer packages having magnetostriction management structures.

**[0007]** One general aspect of the present disclosure includes a transformer based integrated circuit (IC) package. The transformer based IC package may include a substrate including a cavity, where the cavity includes an aperture; a magnetic core disposed in the cavity, where the magnetic core includes a soft ferromagnetic material, where the cavity is configured to provide a space between an interior surface of the cavity and an exterior surface of the magnetic core; a cap disposed in the aperture and configured to seal the aperture; a plurality of conductive traces forming first and second coils disposed about the magnetic core, where the first and second coils and magnetic core are configured as a transformer; and an encapsulant material configured to encapsulate a surface of the substrate and/or magnetic core, where the molding material is configured to form a surface of a package body.

**[0008]** Implementations may include one or more of the following features. The space provided by the cavity may include a gap between the interior surface of the cavity and the exterior surface of the magnetic core. The gap may be between about 250 nm to about 2 mm. The IC package may include at least one semiconductor die disposed on the substrate. The at least one semiconductor die may include an integrated circuit (IC). The IC may include a gate driver. In some embodiments, the first and second coils can be configured as primary and secondary coils in a step-up configuration, step-down, or power transformer configuration. The magnetic core may include ferrite. The substrate may include a printed circuit board (PCB). The cap may include soft ferromagnetic material. The cap may include ferrite. The cap may include mold material. The cap may include a plurality of coil portions of the first and second coils. The substrate can include a step at the aperture, where the step is configured to receive the cap.

**[0009]** Another general aspect of the present disclosure includes a method of making an integrated circuit (IC) and transformer package. The method can include providing a substrate including a cavity, where the cavity includes an aperture; providing a magnetic core disposed in the cavity, where the magnetic core includes a soft ferromagnetic material, where the cavity is configured to provide a space between an interior surface of the cavity and an exterior surface of the magnetic core; providing a cap disposed in the aperture and configured to seal the aperture; providing first and second coils disposed about the magnetic core, where the first and second coils and magnetic core are configured as a transformer; and providing a molding material encapsulating a surface of the substrate, the cap, and the transformer, where the molding material forms a package body.

**[0010]** Implementations may include one or more of the following features. Providing the first and second coils disposed about the magnetic core may include connecting a first plurality of coil portions disposed in the substrate with a second plurality of coil portions disposed exterior to the substrate. The second plurality of coil portions can be provided after the cap is disposed in the aperture. The second plurality of coil portions may include wire bonds. The substrate may include a printed circuit board (PCB). The magnetic core and/or cap may include ferrite. The cap may include mold material. The cap may include a plurality of coil portions of the first and second coils. The substrate may include a step at the aperture, where the step is configured to receive the cap. The method may include providing one or more semiconductor die supported by the substrate. The one or more semiconductor die may include first and second semiconductor die, which may include first and second integrated circuits (ICs), respectively, and the first and second coils may be connected to the first and second ICs, respectively. The first and second ICs may be galvanically isolated. The second IC may include a gate driver.

**[0011]** Another general aspect of the present disclosure includes a voltage-isolated integrated circuit (IC) package. The voltage-isolated IC can include a package body including molding material; a substrate disposed within the package body, where the substrate includes a cavity having an aperture; a magnetic core disposed in the cavity, where the magnetic core includes a soft ferromagnetic material; where the cavity is configured to provide a space between an interior surface of the cavity and an exterior surface of the magnetic core, and where the magnetic core is unconstrained by the substrate; a cap disposed in the aperture and configured to seal the aperture; a plurality of conductive traces forming first and second coils disposed about the magnetic core, where the first and second coils and magnetic core are configured as a transformer; first and second IC die connected to the first and second coils, respectively; a first plurality of leads connected to the first IC die, where the first plurality of leads includes exposed lead portions accessible from the exterior of the package body; and a second plurality of leads connected to the second IC die, where the second plurality of leads includes exposed lead portions accessible from the exterior of the package body.

**[0012]** Implementations may include one or more of the following features. The second IC die of the IC package may include a gate driver. The transformer may be configured as a step-up transformer, a step-down transformer, or a power transformer. The magnetic core and/or cap can include ferrite. The substrate may include a printed circuit board (PCB). The space may include or define a gap between the interior surface of the cavity and the exterior surface of the magnetic core. The gap can be between about 100 nm to about 2 mm in some embodiments. The cap may include a mold material. The cap may include a plurality of coil portions of the first and second coils. The substrate may include a step at the aperture, where the step is configured to receive the cap. The IC package may include a seal disposed between the cap and the substrate. The seal may include epoxy.

**[0013]** The features and advantages described herein are not all-inclusive; many additional features and advantages will be apparent to one of ordinary skill in the art in view of the drawings, specification, and claims. Moreover, it should be noted that the language used in the specification has been selected principally for readability and instructional purposes, and not to limit in any way the scope of the present disclosure, which is susceptible of many embodiments. What follows is illustrative, but not exhaustive, of the scope of the present disclosure.

## BRIEF DESCRIPTION OF THE DRAWINGS

**[0014]** The manner and process of making and using the disclosed embodiments may be appreciated by reference to the figures of the accompanying drawings. In the figures like reference characters refer to like components, parts, elements, or steps/actions; however, similar components, parts, elements, and steps/actions may be referenced by different reference characters in different figures. It should be appreciated that the components and structures illustrated in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principals of the concepts described herein. Furthermore, embodiments are illustrated by way of example and not limitation in the figures, in which:

**[0015]** FIG. **1** includes views (A)-(D) showing side cross sectional, exploded, partially assembled, and assembled views of an example isolation transformer package with magnetostriction management structure, respectively, in accordance with the present disclosure;

**[0016]** FIG. **2** includes views (A)-(D) showing side cross sectional, exploded, partially assembled, and assembled views of an example isolation transformer package including a clip assembly forming coil portions, respectively, in accordance with the present disclosure;

**[0017]** FIG. **3** includes views (A)-(D) showing views of an isolation transformer package a C-core and magnetostriction management structure, in accordance with another embodiment of the present disclosure;

**[0018]** FIG. **4** is a diagram showing an example method of fabricating an isolation transformer package with magnetostriction management structure, in accordance with the present disclosure;

**[0019]** FIGS. **5**A-**5**B show side cross section views of example core and substrate cavity configurations, in accordance with the present disclosure; and

**[0020]** FIG. **6** shows an exploded view of another example of an isolation transformer package with included IC die, in accordance with the present disclosure.

## DETAILED DESCRIPTION

**[0021]** The features and advantages described herein are not all-inclusive; many additional features and advantages will be apparent to one of ordinary skill in the art in view of the drawings, specification, and claims. Moreover, it should be noted that the language used in the specification has been selected principally for readability and instructional purposes, and not to limit in any way the scope of the inventive subject matter. The subject technology is susceptible of many embodiments. What follows is illustrative, but not exhaustive, of the scope of the subject technology.

**[0022]** Aspects of the present disclosure are directed to and include systems, structures, circuits, and methods providing transformers and transformer structures that can be used for galvanic isolation (a.k.a., voltage isolation). Embodiments and examples can include leadframe-based packages with integrated IC-transformer structures having a core and coils in a transformer configuration and providing galvanic isolation for included IC die. In some embodiments, a transformer may have, e.g., a step up, a step down, or a power transformer configuration. Signals and/or power may be transferred from the primary side of the transformer to the secondary side.

**[0023]** One or more (e.g., first and second) semiconductor die having one or more integrated circuits (a.k.a., "IC die") can be included in the transformer packages. Such integrated circuits can include, e.g., but are not limited to, high-voltage circuits such as gate drivers configured to drive an external gate on a solid-state switch, e.g., a field effect transistor (FET), a metal oxide semiconductor FET (MOSFET), a metal semiconductor FET (MESFET), a gallium nitride FET (GaN FET), a high electron mobility transistor (HEMT), a silicon carbide FET (SIC FET), an insulated gate bipolar transistor (IGBT), or another load. The included transformer structure provides galvanic separation between the primary and secondary transformers sides, including any connected ICs.

**[0024]** FIG. **1** includes views (A)-(D) showing side cross sectional, exploded, partially assembled, and assembled views of an example isolation transformer package **100** with magnetostriction management structure, respectively, in accordance with the present disclosure.

**[0025]** View (A) shows package **100** including a substrate **101** with opposed first and second surfaces (sides) **102**, **103**. Substrate **101** can include a laminated structure in some embodiments. Substrate **101** can include a plurality of conductive structures or traces **104**, which may be on a surface of substrate, e.g., surface **102**, and/or included in the interior of substrate **101**. The plurality of conductive traces

**104** can include a first group **104***a* and a second group **104***b* that are galvanically separate from one another. The first group **104***a* can include a plurality of exposed portions (not shown) that are exposed at a first area (or areas) of the substrate **101**, e.g., solder contacts or pads on surface **103**, and the second group can include a plurality of exposed portions (not shown) that are exposed at a second area (or areas) of the substrate **101**, e.g., solder contacts (pads) on surface **103**; the exposed portions can be used for input/output (I/O) functionality for package **100**.

**[0026]** Package **100** also includes transformer **120** having magnetic core **121**, and primary coil **122** and secondary coil **124**, which are configured about transformer core **121** and galvanically separated. Primary coil **122** and secondary coil **124** can each include multiple portions. For example, coils **122** and **124** can include portions exterior to substrate **101** and other portions interior to (e.g., formed or embedded in) substrate **101**, as shown by coils portions **122***a* and **124***a* and **122***b* and **124***b*, respectively. The first and second groups of traces **104***a-b* can be connected to galvanically separated primary and secondary sides of transformer **120**, respectively, as explained in further detail below. Substrate **101** includes a recess (a.k.a., cavity or aperture) **105** defined by walls (surfaces) **106***a-c* for placement of magnetic core **121**. Recess/cavity **105** can form or present an aperture **107** at substrate surface **102** and/or at a surface presented by stepped surface (a.k.a., step) **116**. Recess/cavity **105** can provide a space or gap about core **121**, which can accommodate magnetostriction effects, including changes in size, on core arising from operation of transformer **120**. In some embodiments, the space or gap may be between about 250 nm to about 2 mm (inclusive of any subrange).

**[0027]** A protective cap **114** may be present to facilitate sealing of core **121** within recess **105**. Cap **114** can function to prevent material ingress into recess **105**, e.g., during one or more molding (overmolding) steps applying molding to/on surface **102** or during operation/use of package **100**. In some embodiments, an optional step **116** may be present in a surface of substrate **101**, e.g., side wall **106***a*, to receive or support a portion of cap **114** or core **121** (see **507***a* in FIG. **5**A). Any suitable material may be used for cap **114**. In some examples, cap **114** may include a molding material. In some embodiments, cap **114** may include ferrite.

**[0028]** In some embodiments, substrate **101** can include a printed circuit board (PCB), e.g., a PCB including FR4, FR5, or other PCB material(s). In some embodiments, substrate **101** can include one or more layers of low-temperature cofired ceramic (LTCC) or high-temperature cofired ceramic (HTCC). In some embodiments, substrate **101** can include an alumina substrate or a glass substrate comprising one or more layers of metal and insulation.

**[0029]** Transformer core **121**—shown with cross-sections **121***a-b*—can include one or more soft (referring to magnetic property) ferromagnetic materials. In some embodiments, core **121** can include ferrite, iron particles, ferrosilicon, nickel, nickel alloys (e.g., iron nickel), and/or the like. In some embodiments, core **121** can include a sintered soft ferromagnetic material. Primary coil **122** and secondary coil **124** can each have a desired number of windings, which may differ from coil to coil. Portions of primary and secondary coils **122** and **124**, i.e., coil portion **122***a* and **124***a*, are shown configured (wound) about core **121** as insulated wire or wire bonds, with ends connected to other coil portions, i.e., coil portions **122***b* and **124***b*, of primary and secondary coils **122** and **124** within or on substrate **101**, completing the windings of each coil **122**, **124** about core **121**. Insulating material (not shown) may be disposed between core **121** and substrate **101**. Coils **122** and **124** can be connected, e.g., by way of conductive traces **104**, to respective sets of connection structures, e.g., solder contacts or pads (not shown).

**[0030]** View (B) shows an exploded perspective view of package **100**. IC die **110** and **112** are shown. In some embodiments, IC die **110** and/or **112** can include quad flat no-lead (QFN) packages. Wire portions of primary coil **122***a* and secondary coil **124***a* are shown. Any suitable conductive material(s) may be used for coils **122** and **124**. In some examples, copper or aluminum may be used for wire portions **122***a* and/or **124***a*. Embedded (buried) coil portions **122***b* and **124***b* of primary and secondary coils **122** and **124** are also indicated. In some examples, copper may be used for embedded coil portions **122***b* and/or **124***b*. Secondary coil **124** is shown having a greater number of coil windings than primary coil **122** as would be the case in a step up transformer configuration. The coils **122** and **124** may have different numbers of coil windings in other embodiments. First and second galvanically separated groups of conductive structures/traces **104***a* and **104***b* (which may be internal to substrate **101** and/or on one or more surfaces of substrate **101**) are also indicated and may be made of any suitable conductive material(s), e.g., copper (as a non-limiting example). Optional sealing structure (e.g., silicone sealant/gel and/or epoxy) **128** is indicated.

**[0031]** View (C) shows an assembled perspective view of package **100**. The transformer configuration of primary and secondary coils **122**, **124** and core **121** and galvanic separation of connecting conductive structures/traces **104***a-b* provide galvanic separation of IC **110** and **112** while allowing transfer of power and/or signals between primary and secondary sides of the transformer (e.g., between IC **110** and IC **112**). In some embodiments, primary and secondary coils **122**, **124** may be configured in a step up transformer configuration with secondary coil **124** being on the higher-voltage side. In some embodiments, IC **112** may include a gate driver, e.g., configured to control a semiconductor power switch.

**[0032]** View (D) shows structure of view (C) with an added layer of encapsulant **130** forming package structure **150**. Encapsulant **130** may be added by one or more application steps, e.g., compression molding. Any suitable material may be used for encapsulant **130**. Examples may include, but are not limited to, one or more molding materials, potting materials, dielectric materials, or the like.

**[0033]** FIG. **2** includes views (A)-(D) showing side cross sectional, exploded, partially assembled, and assembled views of an example isolation transformer package **200** including a clip assembly forming coil portions, respectively, in accordance with the present disclosure. Package **200** includes magnetostriction managements structures similar to package **100** of FIG. **1** while also including the clip assembly that forms portions of primary and secondary transformer coils.

**[0034]** View (A) shows package **200** including a substrate **201** with opposed first and second surfaces (sides) **202**, **203**. Substrate **201** can include a laminated structure in some embodiments. Substrate **201** can include a plurality of conductive structures or traces **204**, which may be on a surface of substrate, e.g., surface **202**, and/or included in the interior of substrate **201**. The plurality of conductive traces

**204** can include a first group **204***a* and a second group **204***b* that are galvanically separate from one another. The first group **204***a* can include a plurality of exposed portions (not shown) that are exposed at a first area (or areas) of the substrate **201**, e.g., solder contacts or pads on surface **203**, and the second group can include a plurality of exposed portions (not shown) that are exposed at a second area (or areas) of the substrate **201**, e.g., solder contacts (pads) on surface **203**; the exposed portions can be used for input/output (I/O) functionality for package **200**.

**[0035]** Package **200** also includes transformer **220** with magnetic core **221**, primary coil **222** and secondary coil **224** configured about transformer core **221** and galvanically separated. Primary coil **222** and secondary coil **224** can each include multiple portions. For example, coils **222** and **224** can include portions exterior to substrate **201** and other portions interior to substrate **201**, as shown by coils portions **222***a* and **224***a* and **222***b* and **224***b*, respectively. The first and second groups of traces **204***a-b* can be connected to galvanically separated primary and secondary sides of transformer **220**, respectively, as explained in further detail below. Substrate **201** includes a recess or cavity **205** defined by walls **206***a-c* for placement of magnetic core **221**. Recess/cavity **205** can form or present an aperture **207** at surface **202** and/or at a surface defined by step **216**.

**[0036]** View A also shows assembly (clip) **213** that includes portions of the primary and secondary coils (portions **222***a* and **224***a*), protective cap **214**, and frame **215**, having frame elements **215***a-c*. Protective cap **214** may facilitate sealing of core **221** within recess **205**, and can function to prevent material ingress into recess **205**, e.g., during one or more molding steps (e.g., an overmolding step) applying molding material(s) to/on surface **202**. In some embodiments, an optional step **216** may be present in a surface of substrate **201**, e.g., side wall **206***a*, to receive or support a portion of cap **214** or core **221** (see **507***a* in FIG. **5**A). Coil portions **222***a* and **224***a* can be held by frame elements **215***a-c*. In some embodiments frame elements **215***a-c* can include or be made from insulative material(s) to facilitate/provide galvanic separation of coils **222** and **224**. Assembly (clip) **213** can be used to simplify, reduce cost, and/or reduce required time for fabrication of structure **200**.

**[0037]** In some embodiments, substrate **201** can include a printed circuit board (PCB), e.g., a PCB including FR4, FR5, or other PCB material(s). In some embodiments, substrate **201** can include one or more layers of low-temperature cofired ceramic (LTCC) or high-temperature cofired ceramic (HTCC). In some embodiments, substrate **201** can include an alumina substrate or a glass substrate comprising one or more layers of metal and insulation.

**[0038]** Transformer core **221**—shown with cross-sections **221***a-b*—can include one or more soft (referring to magnetic property) ferromagnetic materials. In some embodiments, core **221** can include ferrite, iron particles, ferrosilicon, nickel, nickel alloys (e.g., iron nickel), and/or the like. In some embodiments, core **221** can include a sintered soft ferromagnetic material. Primary coil **222** and secondary coil **224** can each have a desired number of windings, which may differ from coil to coil. Portions of primary and secondary coils **222** and **224**, i.e., coil portion **222***a* and **224***a*, are shown configured (wound) about core **221** as insulated wire or wire bonds, with ends connected to other coil portions, i.e., coil portions **222***b* and **224***b*, of primary and secondary coils **222** and **224** within or on substrate **201**, completing the windings of each coil **222**, **224** about core **221**. Insulating material (not shown) may be disposed between core **221** and substrate **201**. Coils **222** and **224** can be connected, e.g., by way of conductive traces **204**, to respective sets of connection structures, e.g., solder contacts or pads (not shown).

**[0039]** View (B) shows an exploded perspective view of package **200**. IC die **210** and **212** are shown. In some embodiments, IC die **210** and/or **212** can include quad flat no-lead (QFN) packages. Wire portions of primary coil **222***a* and secondary coil **224***a* are shown. Embedded coils portions **222***b* and **224***b* of primary and secondary coils **222** and **224** are indicated. Secondary coil **224** is shown having a greater number of coil windings than primary coil **222** as would be the case in a step up transformer configuration. The coils may have different numbers of coil windings in other embodiments. First and second galvanically separated groups of conductive structures/traces **204***a* and **204***b* (which may be internal to substrate **201** and/or on one or more surfaces of substrate **201**) are also indicated. Assembly (clip) **213**, with coil portions **222***a* and **224***a*, is shown with protective cap **214** and frame **215** (with frame elements **215**a-c). Optional sealing structure (e.g., silicone sealant/gel and/or epoxy) **228** is indicated.

**[0040]** View (C) shows an assembled perspective view of package **200**. The transformer configuration of primary and secondary coils **222**, **224** and core **221** and galvanic separation of connecting conductive structures/traces **204***a-b* provide galvanic separation of IC **210** and **212** while allowing transfer of power and/or signals between primary and secondary sides of the transformer (e.g., between IC **210** and IC **212**). In some embodiments, primary and secondary coils **222**, **224** may be configured in a step up transformer configuration with secondary coil **224** being on the higher-voltage side.

**[0041]** View (D) shows structure of view (C) with an added layer of encapsulant **230** forming package structure **250**. Encapsulant **230** may be added by one or more application steps, e.g., compression molding. Any suitable material may be used for encapsulant **230**. Examples may include, but are not limited to, one or more molding materials, potting materials, dielectric materials, or the like.

**[0042]** Some examples and embodiments of the present disclosure can include magnetic cores that are not necessarily configured entirely in a recess in the substrate and generally parallel to the substrate. Some examples and embodiments can include cores having portions (such as legs or plates or other similar structures) that pass through one or more apertures in a substrate, e.g., so-called C-cores or E-cores.

**[0043]** FIG. **3** includes views (A)-(D) showing views of an isolation transformer package **300** with a C-core and magnetostriction management structure, in accordance with another embodiment of the present disclosure. As shown, package **300** includes a so-called C-core as a magnetic core for the included transformer.

**[0044]** As shown in view (A), package **300** includes substrate **301** having opposed first and second surfaces (sides) **302**, **303**. Substrate **301** can include conductive structures and/or traces **304** interior to substrate **301** and/or disposed on one or more surfaces **302**, **303** for connections to package components and/or as I/O structures. Conductive structures **304** can include first and second groups **304***a*, **304***b* that are galvanically separated. Primary die **310** and secondary die **312** are shown disposed on surface **302**. In some embodiments, IC die **310** and/or **312** can include quad flat no-lead (QFN) packages. Transformer **320** includes magnetic core **321** and primary and secondary coils **322** and **324**. Coils **322** and **324** may be entirely embedded or disposed within substrate **301**. Substrate **301** can include one or more recesses (e.g., recesses **306**, **307**) on surfaces **302** and/or **303** to receive core **321**. The transformer configuration of primary and secondary coils **322**, **324** and core **321** can provide galvanic separation of IC **310** and **312** while allowing transfer of power and/or signals between primary and secondary sides of the transformer (e.g., between IC **310** and IC **312**). In some embodiments, primary and secondary coils **322**, **324** may be configured in a step up transformer configuration with secondary coil **324** being on the higher-voltage side.

**[0045]** Magnetic core **321** can include any suitable soft ferromagnetic material(s) and may be configured as a C-core with multiple portions that can be connected together for operation of transformer **320**. For example, three core portions **321***b-d* may formed/connected together and core portion **321***a* may be coupled to those joined/integral portions after they have been positioned for use relative to substrate **301**.

**[0046]** As shown in view (B), substrate **301** can include recess **307** in surface **303**. Recess **307** can be configured to receive a portion of core **321**, e.g., a bottom plate of core **321**. In some embodiments, recess **307** can have a shape that matches that of the received portion of core **321**. Substrate **301** can also include one or more apertures **308** that are configured to receive one or more portions of core **321**, e.g., cylindrical core portions (a.k.a., legs or arms) **321***c-d*. Apertures **308** can pass from surface **302** to **303**.

**[0047]** As shown in view (C), substrate **301** can include recess **306** in surface **302**, with recess **306** being configured to receive a portion of core **321** e.g., a top plate of core **321**. In some embodiments, recess **307** can have a shape that matches that of the received portion of core **321**. In some embodiments, a seal structure or material (sealant) may be applied between the received portion(s) of core **321** and substrate **301** to facilitate prevention of material ingress to recesses **306**, **307** and/or apertures **308** during fabrication, e.g., an overmolding step, and/or during operation of transformer **320**.

**[0048]** As shown in view (D), an encapsulant material **330** can be applied to substrate **301**, e.g., to surface **302**. Any suitable material(s) may be used for encapsulant **330**. Examples may include, but are not limited to, molding materials, potting materials, dielectric materials, etc. Encapsulant **330** may define one or more surfaces of a package structure or body **350**.

**[0049]** FIG. **4** is a diagram showing an example method of fabricating an isolation transformer package with magnetostriction management structure, in accordance with the present disclosure. Method **400** can include providing a substrate including a cavity or recess, with the cavity including one or more apertures, as described at **402**. A magnetic core can be provided that is disposed in the cavity or recess, with the magnetic core including one or more soft ferromagnetic materials, and with the cavity providing a space (gap) around the magnetic core, as described at **404**. Method **400** can include providing a cap covering the recess and/or disposed in the aperture(s) and configured to seal the aperture(s), as described at **406**. In some embodiments, the cap can include a portions of the magnetic core, e.g., when the magnetic core is configured as a C-core.

**[0050]** First and second coils can be provided that are disposed about the magnetic core, with the first and second coils and magnetic core being configured as a transformer, as described at **408**. Method **400** can optionally include providing an encapsulant (encapsulate) material, encapsulating at least a portion of the substrate and/or the cap. The encapsulant material can define one or more surfaces of a package body. In some embodiments, the encapsulant can include a molding material, a potting material, and/or a dielectric material.

**[0051]** FIGS. **5**A-**5**B show side cross section views of example core and substrate cavity configurations **500**A, **500**B, in accordance with the present disclosure.

**[0052]** FIG. **5**A shows substrate **501** having first and second opposed surfaces (sides) **502**, **503**. Substrate **501** can include a plurality of conductive structures or traces **504**, which may be interior to the substrate **501** and/or disposed on one or more surfaces of substrate **501**, e.g., surface **502**, surface **503**. Conductive structures/traces **504** can include first and second groups **504***a-b* that are galvanically separated. Conductive structures/traces **504** can connect components mounted to or included in substrate **501** and/or form I/O structures. Substrate **501** can include a recess or cavity **505**, which may receive magnetic core **506**. Cavity **505** can include an outer step **505***a*, an inner step **505***b*, outer sidewall **505***c*, and inner sidewall **505***d*, as shown. Core **506** may include one or more soft ferromagnetic materials (e.g., ferrite) and may have a closed shape, as indicated by cross sections **506***a-b*, each shown with sidewalls **506***c-d*. A portion of substrate **501**, i.e., portion **501***a*, e.g., an interior portion or central island, is shown surrounded by recess **505**. Recess/cavity **505** can provide a space or gap about core **506**; the space or gap provided by cavity **505** can accommodate magnetostriction effects, including changes in size, on core **506** arising from operation of transformer **120**. In some embodiments, the space or gap may be between about 250 nm to about 2 mm (inclusive of any subrange). The space or gap is not necessarily uniform about core **506** and in some embodiments (see FIG. **5**B), the gap (space) may vary.

**[0053]** As shown, core **506** may have a T-shaped cross section that includes a wider portion **507** that has overhangs or lips—shown as lip/overhang **507***a* and lip/overhang **507***b*—which can be supported by steps **505***a* and **505***b*, respectively, when core **506** is disposed in recess/cavity **505**. This configuration of core **506**, with portion **507**, can enable the core (e.g., ferrite) to create a seal (e.g., at points **506***e*) even after accommodating the manufacturing tolerances of the core and substrate **501** (e.g., PCB).

**[0054]** FIG. **5**B shows substrate **501** (similar to as shown in FIG. **5**A) receiving core **506**, which has a tapered or wedge-shaped cross section (shown by **506***a-b*). Core **506** may have a closed shape, e.g., one that is oval or rectangular, and is disposed in cavity/recess **505** having stepped surfaces **505***a-b* and sidewalls **505***c-d*. Stepped surfaces **505***a-b* are optional and may be omitted in some embodiments.

**[0055]** As shown, core **506** has sidewalls **506***c-d* that are non-parallel relative to cavity sidewalls **505***c-d*, respectively. The core sidewalls **506***c-d* and the cavity sidewalls **505***c-d* can accordingly form an angle (a.k.a., a draft angle), shown as **509**. The degree of magnitude of the draft angle **509** may be implemented as desired. A seal, between the core **506** and the substrate at the perimeters of the cavity, can be created with the draft angle on the mating edges of a core **506**, as shown. Sealing the recess/cavity **505** with a wedge shape provided by the core **506** may, in some situations, result in the core (e.g., at cross sections shown) sitting at different heights within the recess/cavity, e.g., due to cores and/or recesses being manufactured with manufacturing variations, even though they may be within design tolerances.

**[0056]** FIG. **6** shows an exploded view of another example of an isolation transformer package **600** with included IC die, in accordance with the present disclosure. Substrate **601** is shown with sidewall (wettable) **601**a and opposed first and second sides (surfaces) **602**, **603**. Substrate **601** includes a plurality of conductive structures/traces **604**, which may be disposed within and/or on substrate **601**. The plurality of conductive structures/traces **604** can include first and second groups that are galvanically separate. Galvanically separated primary and secondary coils **622** and **624** include first (wire) portions **622***a*, **624***a* and second (buried) portions **622***b*, **624***b* respectively. Coil portions **622***a* and **624***a* may have a designed/desired separation distance, d, as shown. Separation distance "d" may be selected as desired, e.g., to meet a given creepage requirement for a specification/rating. While core **621** is shown removed from substrate **601**, core **621** would be disposed in cavity/recess **605** prior to wire portions **622***a*, **624***a* being connected. Core **621** can include overhang or lips **621***a-b*, which can be received by or disposed on step **606***a* formed in recess sidewall **606**. An encapsulation layer **630** of encapsulant material can be included. Any suitable encapsulant may be used for layer **630**. Suitable encapsulant materials may include, but are not limited to, one or more molding materials, potting materials, dielectric materials, and/or the like. While not shown, other components such as ID die can be connected to the primary and secondary sides of transformer **620** within package **600**.

**[0057]** In some examples and/or embodiments, integrated circuits (ICs), e.g., in IC die **310** and **312** in FIG. **3**, or other conductive features of the primary and secondary sides of a transformer structure in an IC or transformer package according to the present disclosure can be fabricated or configured to have a desired separation distance (d) between certain parts or features, e.g., to meet internal creepage or external clearance requirements for a given pollution degree rating as defined by certain safety standards bodies such as the Underwriters Laboratories (UL) and the International Electrotechnical Commission (IEC). For example, a separation distance may be between closest (voltage) points of the respective circuits, e.g., the low-voltage (primary) side and high-voltage (secondary) side. For further example, such a separation distance may be the distance between any two voltage points between the primary and secondary sides, e.g., a distance between die, or a distance between exposed leads connected to the die, may be, may be approximately, or may be at least 1.2 mm, 1.4 mm, 1.5 mm, 3.0 mm, 4.0 mm, 5.5 mm, 7.2 mm, 8.0 mm, 10 mm, or 10+mm in respective examples. Such a distance between conductive portions or areas of die can include any insulation covering a conductor, e.g., such as plastic coating of a wire/lead. Other distances between conductive parts, components, and/or features of an IC/transformer package may also be designed and implemented, e.g., to meet desired internal creepage, voltage breakdown, or external clearance requirements, e.g., between external leads.

**[0058]** In some examples and embodiments, a dielectric material (e.g., gel) may be used for potting and/or protecting substrate systems, assemblies, and/or packages, including magnetic/or die and/or interconnects from environment conditions and/or to provide dielectric insulation. In some examples, a dielectric material may include, but is not limited to, one or more of the following materials: DOWSIL™ EG-3810 Dielectric Gel (made available by The Dow Chemical Corporation, a.k.a., "Dow"), and DOWSIL™ EG-3896 Dielectric Gel (made available by Dow), which has the ability to provide isolation greater than 20 kV/mm. Other suitable gel materials may also or instead be used, e.g., to meet or facilitate meeting/achieving voltage isolation specifications required by a given package design. DOWSIL™ EG-3810 is designed for temperature ranges from −60° C. to 200° C. and DOWSIL™ EG-3896 Dielectric Gel −40° C. to +185° C.; both of which can be used to meet typical temperature ranges for automotive applications.

**[0059]** Accordingly, embodiments and/or examples of the inventive subject matter can afford various benefits relative to prior art techniques. For example, embodiments and examples of the present disclosure can enable or facilitate use of smaller size packages for a given power, current. or voltage rating. Embodiments and examples of the present disclosure can enable or facilitate lower costs and higher scalability for manufacturing of IC packages/modules having voltage-isolated (galvanic isolation) IC die and transformers.

**[0060]** Various embodiments and/or examples of the concepts, systems, devices, structures, and techniques sought to be protected are described above with reference to the related drawings. Alternative embodiments can be devised without departing from the scope of the concepts, systems, devices, structures, and techniques described.

**[0061]** It is noted that various connections and positional relationships (e.g., over, below, adjacent, etc.) may be used to describe elements and components in the description and drawings. These connections and/or positional relationships, unless specified otherwise, can be direct or indirect, and the described concepts, systems, devices, structures, and techniques are not intended to be limiting in this respect. Accordingly, a coupling of entities can refer to either a direct or an indirect coupling, and a positional relationship between entities can be a direct or indirect positional relationship.

**[0062]** As an example of an indirect positional relationship, positioning element "A" over element "B" can include situations in which one or more intermediate elements (e.g., element "C") is between elements "A" and elements "B" as long as the relevant characteristics and functionalities of elements "A" and "B" are not substantially changed by the intermediate element(s).

**[0063]** Also, the following definitions and abbreviations are to be used for the interpretation of the claims and the specification. The terms "comprise," "comprises," "comprising," "include," "includes," "including," "has," "having," "contains" or "containing," or any other variation are intended to cover a non-exclusive inclusion. For example, an apparatus, a method, a composition, a mixture, or an article, which includes a list of elements is not necessarily limited to only those elements but can include other elements not expressly listed or inherent to such apparatus, method, composition, mixture, or article.

**[0064]** Additionally, the term "exemplary" means "serving as an example, instance, or illustration." Any embodiment or design described as "exemplary" is not necessarily to be construed as preferred or advantageous over other embodiments or designs. The terms "one or more," "plurality," and "at least one" indicate any integer number greater than or equal to one, i.e., one, two, three, four, etc. Those terms, however, may refer to fractional numbers/values greater than one where context admits, e.g., "one or more windings," "a plurality" of windings or "at least one winding" can refer to a number of windings of a coil having a fractional value such as 1.5, 2.75, 3.8, 6.6, etc. The term "connection" can include an indirect connection and a direct connection.

**[0065]** References in the specification to "embodiments," "one embodiment, "an embodiment," "an example embodiment," "an example," "an instance," "an aspect," etc., indicate that the embodiment described can include a particular feature, structure, or characteristic, but every embodiment may or may not include the particular feature, structure, or characteristic. Moreover, such phrases do not necessarily refer to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it may affect such feature, structure, or characteristic in other embodiments whether explicitly described or not.

**[0066]** Relative or positional terms including, but not limited to, the terms "upper," "lower," "right," "left," "vertical," "horizontal, "top," "bottom," and derivatives of those terms relate to the described structures and methods as oriented in the drawing figures. The terms "overlying," "atop," "on top, "positioned on" or "positioned atop" mean that a first element, such as a first structure, is present on a second element, such as a second structure, where intervening elements such as an interface structure can be present between the first element and the second element. The term "direct contact" means that a first element, such as a first structure, and a second element, such as a second structure, are connected without any intermediary elements.

**[0067]** Use of ordinal terms such as "first," "second," "third," etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another, or a temporal order in which acts of a method are performed but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish the claim elements.

**[0068]** The terms "approximately" and "about" may be used to mean within ±20% of a target (or nominal) value in some embodiments, within plus or minus (±) 10% of a target value in some embodiments, within ±5% of a target value in some embodiments, and yet within ±2% of a target value in some embodiments. The terms "approximately" and "about" may include the target value. The term "substantially equal" may be used to refer to values that are within ±20% of one another in some embodiments, within ±10% of one another in some embodiments, within ±5% of one another in some embodiments, and yet within ±2% of one another in some embodiments.

**[0069]** The term "substantially" may be used to refer to values that are within ±20% of a comparative measure in some embodiments, within ±10% in some embodiments, within ±5% in some embodiments, and yet within ±2% in some embodiments. For example, a first direction that is "substantially" perpendicular to a second direction may refer to a first direction that is within ±20% of making a 90° angle with the second direction in some embodiments, within ±10% of making a 90° angle with the second direction in some embodiments, within ±5% of making a 90° angle with the second direction in some embodiments, and yet within ±2% of making a 90° angle with the second direction in some embodiments.

**[0070]** The disclosed subject matter is not limited in its application to the details of construction and to the arrangements of the components set forth in the following description or illustrated in the drawings. The disclosed subject matter is capable of other embodiments and of being practiced and carried out in various ways.

**[0071]** Also, the phraseology and terminology used in this patent are for the purpose of description and should not be regarded as limiting. As such, the conception upon which this disclosure is based may readily be utilized as a basis for the designing of other structures, methods, and systems for carrying out the several purposes of the disclosed subject matter. Therefore, the claims should be regarded as including such equivalent constructions as far as they do not depart from the spirit and scope of the disclosed subject matter.

**[0072]** Although the disclosed subject matter has been described and illustrated in the foregoing exemplary embodiments, the present disclosure has been made only by way of example. Thus, numerous changes in the details of implementation of the disclosed subject matter may be made without departing from the spirit and scope of the disclosed subject matter.

**[0073]** Accordingly, the scope of this patent should not be limited to the described implementations but rather should be limited only by the spirit and scope of the following claims.

**[0074]** All publications and references cited in this patent are expressly incorporated by reference in their entirety.

What is claimed is:

**1**. A transformer based integrated circuit (IC) package comprising:

- a substrate including a cavity, wherein the cavity includes an aperture;
- a magnetic core disposed in the cavity, wherein the magnetic core includes a soft ferromagnetic material, wherein the cavity is configured to provide a space between an interior surface of the cavity and an exterior surface of the magnetic core;
- a cap disposed in the aperture and configured to seal the aperture;
- a plurality of conductive traces forming first and second coils disposed about the magnetic core, wherein the first and second coils and magnetic core are configured as a transformer; and
- an encapsulant material configured to encapsulate a surface of the substrate and/or magnetic core, wherein the molding material is configured to form a surface of a package body.

**2**. The IC package of claim **1**, wherein the space comprises a gap between the interior surface of the cavity and the exterior surface of the magnetic core.

**3**. The IC package of claim **2**, wherein the gap is between about **250** nm to about **2** mm.

**4**. The IC package of claim **1**, further comprising at least one semiconductor die disposed on the substrate.

**5**. The IC package of claim **4**, wherein the at least one semiconductor die comprises an integrated circuit (IC).

**6**. The IC package of claim **5**, wherein the IC comprises a gate driver.

**7**. The IC package of claim **6**, wherein the first and second coils are configured as primary and secondary coils in a step-up configuration.

**8**. The IC package of claim **1**, wherein the magnetic core comprise ferrite.

**9**. The IC package of claim **1**, wherein the substrate comprises a printed circuit board (PCB).

**10**. The IC package of claim **1**, wherein the cap comprises soft ferromagnetic material.

**11**. The IC package of claim **10**, wherein the cap comprises ferrite.

**12**. The IC package of claim **1**, wherein the cap comprises mold material.

**13**. The IC package of claim **12**, wherein the cap comprises a plurality of coil portions of the first and second coils.

**14**. The IC package of claim **1**, wherein the substrate includes a step at the aperture, wherein the step is configured to receive the cap.

**15**. A method of making an integrated circuit (IC) and transformer package, the method comprising:

providing a substrate including a cavity, wherein the cavity includes an aperture;

providing a magnetic core disposed in the cavity, wherein the magnetic core includes a soft ferromagnetic material, wherein the cavity is configured to provide a space between an interior surface of the cavity and an exterior surface of the magnetic core;

providing a cap disposed in the aperture and configured to seal the aperture;

providing first and second coils disposed about the magnetic core, wherein the first and second coils and magnetic core are configured as a transformer; and

providing a molding material encapsulating a surface of the substrate, the cap, and the transformer, wherein the molding material forms a package body.

**16**. The method of claim **15**, wherein providing the first and second coils disposed about the magnetic core comprises connecting a first plurality of coil portions disposed in the substrate with a second plurality of coil portions disposed exterior to the substrate.

**17**. The method of claim **16**, wherein the second plurality of coil portions are provided after the cap is disposed in the aperture.

**18**. The method of claim **16**, wherein the second plurality of coil portions comprises wire bonds.

**19**. The method of claim **15** wherein the substrate comprises a printed circuit board (PCB).

**20**. The method of claim **15**, wherein the magnetic core and/or cap comprise ferrite.

**21**. The method of claim **15**, wherein the cap comprises mold material.

**22**. The method of claim **21**, wherein the cap comprises a plurality of coil portions of the first and second coils.

**23**. The method of claim **15**, wherein the substrate includes a step at the aperture, wherein the step is configured to receive the cap.

**24**. The method of claim **15**, further comprising providing one or more semiconductor die supported by the substrate.

**25**. The method of claim **24**, wherein the one or more semiconductor die comprise first and second semiconductor die comprising first and second integrated circuits (ICs), respectively, and wherein the first and second coils are connected to the first and second ICs, respectively.

**26**. The method of claim **25**, wherein the first and second ICs are galvanically isolated.

**27**. The method of claim **25**, wherein the second IC comprises a gate driver.

**28**. A voltage-isolated integrated circuit (IC) package comprising:

a package body including molding material;

a substrate disposed within the package body, wherein the substrate includes a cavity having an aperture;

a magnetic core disposed in the cavity, wherein the magnetic core includes a soft ferromagnetic material;

wherein the cavity is configured to provide a space between an interior surface of the cavity and an exterior surface of the magnetic core, and wherein the magnetic core is unconstrained by the substrate;

a cap disposed in the aperture and configured to seal the aperture;

a plurality of conductive traces forming first and second coils disposed about the magnetic core, wherein the first and second coils and magnetic core are configured as a transformer;

first and second IC die connected to the first and second coils, respectively;

a first plurality of leads connected to the first IC die, wherein the first plurality of leads includes exposed lead portions accessible from the exterior of the package body;

and a second plurality of leads connected to the second IC die, wherein the second plurality of leads includes exposed lead portions accessible from the exterior of the package body.

**29**. The IC package of claim **28**, wherein the second IC die comprises a gate driver.

**30**. The IC package of claim **28**, wherein the transformer is configured as a step-up transformer.

**31**. The IC package of claim **28**, wherein the magnetic core and/or cap includes ferrite.

**32**. The IC package of claim **28**, wherein the substrate comprises a printed circuit board (PCB).

**33**. The IC package of claim **28**, wherein the space comprises a gap between the interior surface of the cavity and the exterior surface of the magnetic core.

**34**. The IC package of claim **33**, wherein the gap is between about **100** nm to about **2** mm.

**35**. The IC package of claim **28**, wherein the cap comprises a mold material.

**36**. The IC package of claim **35**, wherein the cap comprises a plurality of coil portions of the first and second coils.

**37**. The IC package of claim **28**, wherein the substrate includes a step at the aperture, wherein the step is configured to receive the cap.

**38**. The IC package of claim **28**, further comprising a seal disposed between the cap and the substrate.

**39**. The IC package of claim **38**, wherein the seal comprises epoxy.

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